

Cyber Security

LEARNING ROADMAP

Fundamental IT Skills

CORE BASICS

Computer Hardware Components
Connection Types & Function
OS-Independent Troubleshooting

POPULAR SUITES

MS Office Suite Google Suite iCloud

BASICS OF NETWORKING

WiFi Bluetooth NFC Infrared

CTFS (CAPTURE THE FLAG)

HackTheBox TryHackMe VulnHub picoCTF
SANS Holiday Hack Challenge

Certifications

BEGINNER CERTIFICATIONS

CompTIA A+ CompTIA Linux+ CompTIA Network+ CCNA CompTIA Security+

ADVANCED CERTIFICATIONS

CEH CISA CISM GSEC GPEN GWAPT GIAC OSCP CREST CISSP

Operating Systems

WINDOWS · LINUX · MACOS

- Installation and Configuration
- Different Versions and Differences
- Navigating using GUI and CLI
- Understand Permissions
- Installing Software and Applications
- Performing CRUD on Files
- Troubleshooting
- Common Commands (learn for each OS)

Networking Knowledge

CORE CONCEPTS

OSI Model Common Protocols & Uses
Common Ports & Uses SSL & TLS Basics
NAS vs SAN Subnetting Basics
Public vs Private IP

IP TERMINOLOGY

localhost loopback CIDR
subnet mask default gateway

TERMINOLOGY

VLAN DMZ ARP VM DHCP
DNS NAT IP Router Switch
VPN MAN LAN WAN WLAN
NTP IPAM

NETWORK TOPOLOGIES

Star Ring Mesh Bus

NETWORK PROTOCOLS

SSH RDP FTP SFTP
HTTP / HTTPS SSL / TLS

VIRTUALIZATION

VMWare VirtualBox ESXi Proxmox Hypervisor VM GuestOS
HostOS

TROUBLESHOOTING TOOLS

nslookup iptables ipconfig netstat ping dig arp route
nmap tcpdump traceroute Packet Sniffers Port Scanners
Protocol Analyzers

AUTHENTICATION METHODOLOGIES

Kerberos RADIUS LDAP SSO Certificates Local Auth

Security Skills and Knowledge

FOUNDATIONAL CONCEPTS

Common Hacking Tools Exploit Frameworks Defense in Depth Runbooks Forensics Basics Threat Hunting Vulnerability Management Reverse Engineering Pentest Rules of Engagement Perimeter vs DMZ vs Segmentation
Zero Trust Role of Compliance & Auditors Definition of Risk Backups & Resiliency Cyber Kill Chain MFA & 2FA OS Hardening Concept of Isolation IDS & IPS Honeyspots Authentication vs Authorization
Blue / Red / Purple Teams False Negative / Positive True Negative / Positive Threat Intel & OSINT Handshakes CIA Triad Privilege Escalation Web Attacks & OWASP Top 10 Malware Types

IR & DISCOVERY TOOLS

nmap traceroute nslookup curl
hping ping arp cat dd head
grep tail dig ipconfig wireshark
winhex memdump FTK Imager
autopsy

CRYPTOGRAPHY

Salting Hashing Key Exchange PKI
Private vs Public Keys Obfuscation

FRAMEWORKS & STANDARDS

ATT&CK Diamond Model Kill Chain
ISO NIST RMF CIS CSF

SIEM / SOAR & DISTROS

SIEM SOAR ParrotOS Kali Linux

USING TOOLS FOR UNINTENDED PURPOSES

LOLBAS GTFOBins WADCOMS

LOGS TO LEARN

Event Logs Syslogs Netflow
Packet Captures Firewall Logs

HARDENING CONCEPTS

MAC-based NAC-based Port Blocking
Group Policy Sinkholes ACLs
Patching Jump Server
Endpoint Security

SECURE VS INSECURE PROTOCOLS

FTP vs SFTP SSL vs TLS IPSEC
DNSSEC LDAPS SRTP S/MIME

SECURITY TERMS

Antivirus Antimalware EDR DLP
ACL Firewall & NGFW HIPS NIDS
NIPS Host Based Firewall Sandboxing
EAP vs PEAP WPS
WPA vs WPA2 vs WPA3 vs WEP

INCIDENT RESPONSE PROCESS

Preparation Identification
Containment Eradication Recovery
Lessons Learned

THREAT CLASSIFICATION

Zero Day Known vs Unknown APT

COMMON ANALYSIS TOOLS

VirusTotal Joe Sandbox any.run urlvoid urlscan WHOIS

ATTACK TYPES & DIFFERENCES

Phishing Vishing Whaling Smishing Spam vs Spim Shoulder Surfing Dumpster Diving Tailgating Zero Day Social Engineering Reconnaissance Impersonation Watering Hole Attack Drive-by Attack Typo Squatting
Brute Force vs Password Spray

COMMON ATTACKS

DoS vs DDoS MITM Spoofing Evil Twin DNS Poisoning Deauth Attack VLAN Hopping Rogue Access Point Buffer Overflow Memory Leak SQL Injection XSS CSRF Replay Attack Pass the Hash Directory Traversal

UNDERSTAND AUDIENCE

Stakeholders HR Legal Compliance Management

Cloud Skills and Knowledge

CORE CONCEPTS

Security in the Cloud General Flow of Cloud Deployment Cloud vs On-Premises Infrastructure as Code Concept of Serverless

CLOUD SERVICES

SaaS PaaS IaaS

CLOUD MODELS

Private Public Hybrid

COMMON CLOUD ENVIRONMENTS

AWS GCP Azure

COMMON CLOUD STORAGE

S3 Dropbox Box OneDrive Google Drive
iCloud

Programming Skills

Python Go JavaScript C++ Bash PowerShell

TheSudoLab

Keep Learning — find the detailed version of this roadmap along with other similar roadmaps at

thesudolab.xyz